

Week 3 - Day 6

Random Forest & Model Comparison

AIM2 Preparation | Status: COMPLETE

Day Goal

Understand ensemble learning and Random Forest, compare classification models fairly, interpret feature importance, and choose evaluation metrics based on business goals.

1. Random Forest Fundamentals

Decision Tree: one tree learns feature-based rules. **Random Forest:** many randomized Decision Trees are trained and their predictions are combined.

Ensemble: multiple models combined into one final prediction.

n_estimators: number of trees in the forest. n_estimators=100 means 100 trees.

Bootstrap sampling: training rows are sampled with replacement, so rows may repeat and some may be absent from an individual tree's sample.

Random feature selection: random subsets of features are considered at splits, encouraging diverse, less-correlated trees.

Different samples + different feature choices -> diverse trees -> combined vote -> final classification.

2. Overfitting and Generalization

Random Forest often reduces the instability and variance of one Decision Tree, but it can still overfit. Always evaluate on unseen data.

3. Day 6 Dataset & Workflow

Step	What was done
1	Created a 60-row synthetic customer churn dataset
2	Inspected data and target balance
3	X = Monthly_Charges, Contract_Months, Support_Calls, Tenure_Months
4	y = Churn
5	Shared stratified 75/25 train-test split
6	Trained Logistic Regression, Decision Tree, and Random Forest
7	Compared Accuracy, Precision, Recall, and F1
8	Inspected Random Forest feature importance
9	Experimented with n_estimators

Target Balance

Stay = 30 (50%); Churn = 30 (50%). The target is perfectly balanced.

Fair comparison: all models use the same train/test split.

4. Correct Metric Evaluation

For classifiers, `model.score(X_test, y_test)` returns accuracy. Precision, Recall, and F1 must be calculated from each model's own predictions.

```
lr_pred = lr_model.predict(X_test)
dt_pred = dt_model.predict(X_test)
rf_pred = rf_model.predict(X_test)

accuracy_score(y_test, rf_pred)
precision_score(y_test, rf_pred)
recall_score(y_test, rf_pred)
f1_score(y_test, rf_pred)
```

5. Final Model Comparison

Model	Accuracy	Precision	Recall	F1
Logistic Regression	1.000	1.000	1.000	1.000
Decision Tree (depth=3)	0.933	0.875	1.000	0.933
Random Forest (100 trees)	1.000	1.000	1.000	1.000

Accuracy: Logistic Regression and Random Forest tie at 1.00.

Precision: Logistic Regression and Random Forest tie at 1.00.

Recall: all three models tie at 1.00.

F1: Logistic Regression and Random Forest tie at 1.00.

Decision Tree Recall=1.00 means it found every actual churner in this test set. Precision=0.875 means some predicted churners were actually stayers.

100% test performance on a small synthetic test set does not prove perfect real-world performance.

6. Business-Driven Model Selection

Missing churners is costly: prioritize Recall (reduce false negatives).

Retention offers are expensive: prioritize Precision (reduce false positives).

Need balance: consider F1.

The best model is not automatically the one with the highest Accuracy; business goals and error costs matter.

7. Random Forest Feature Importance

Feature	Importance
Tenure_Months	0.4220
Monthly_Charges	0.3655
Support_Calls	0.1812
Contract_Months	0.0313

The fitted forest relied most on **Tenure_Months**. Feature importance describes predictive contribution in this model; it does not establish causation.

Feature importance != causation

8. n_estimators Experiment

Trees	Training Accuracy	Testing Accuracy
10	1.000	0.933
50	1.000	1.000
100	1.000	1.000
200	1.000	1.000

Test accuracy improved from 10 to 50 trees, then plateaued at 1.00 for 100 and 200 trees.

More trees do not guarantee continuously improving accuracy; returns can diminish while computation increases.

9. Assessment - Key Answers

Q1: Ensemble = multiple models combined. **Q2:** Decision Tree = one tree; Random Forest = many randomized trees.

Q3: n_estimators=100 means 100 trees. **Q4:** Bootstrap = sample rows with replacement.

Q5: Random feature subsets increase diversity. **Q6:** Random Forest can still overfit.

Q7: Better unseen performance and a smaller train/test gap suggest better generalization.

Q8: Use the same split for fair comparison. **Q9:** Prioritize Recall when missing churners matters most.

Q10: Choose the higher-Recall model when false negatives are much more costly.

10. Quick Revision Sheet

Concept	Remember
Ensemble	Multiple models combined
Random Forest	Many randomized Decision Trees
Bootstrap	Sample rows with replacement
Random features	Increase tree diversity
n_estimators	Number of trees
Accuracy	Overall correctness
Precision	Trust positive predictions; false positives matter
Recall	Find actual positives; false negatives matter
F1	Balance Precision and Recall
Feature importance	Predictive contribution, not causation
Fair comparison	Same train/test split
Model selection	Business goal, not Accuracy alone

Day 6 Completion

Status: COMPLETE

Week 3 progress: **6/7 days complete.**

Next: Day 7 - Customer Churn Prediction final project.